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Emulation Digital Twin Framework: Case Study on NGAP Attacks in 5G Control Plane

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***Abstract*—The signalling protocols at the 5G network control plane may be exploited to perform Denial-of-Service (DoS) attacks. The countermeasures developed have to be evaluated in realistic conditions to validate their effectiveness and performance overhead. However, current emulation frameworks offer limited automation, which prevents large-scale and iterative what-if analysis. To address these needs, we present the Emulation Digital Twin (EDT) Framework, which is a modular three-layered architecture. Here, the Emulation Layer utilises open-source tools of Containerlab, UERANSIM, and an extended Open5GS with designed signalling hooks. Then, the Digital Twin (DT) Layer utilises these to form a graph-based representation and employs the attack detection and mitigation modules. Furthermore, the Orchestration Layer automates scenario generation, deployment, traffic/attack injection, and experiment management. We validate EDT, respectively, with a case study on Next Generation Application Protocol (NGAP) DoS attack performed through the N2 interface between a gNB and Access and Mobility Function (AMF). Here, we showcase how the EDT framework reduces experimental configuration complexity by 81% in lines of code with minimal resource consumption increase.**

***Index Terms*—digital twin, 5G/6G, network emulation, attack detection, what-if**

I. Introduction

6G networks will be flexible and scalable thanks to being fully virtualised with distributed compute resources across radio access, edge, and core. However, these also expand and complicate the attack surface [1]. For example, a Denial-of-Service (DoS) attack on a Network Function (NF) will also affect the other NFs deployed on that physical machine. While a level of isolation is present between the co-located NFs, this is not absolute as they share resources at the host level, such as memory buses, storage I/O, and network interfaces [2]. Moreover, 5G operates through control plane functions that coordinate to provide services. Therefore, a disruption to an NF will propagate to the wider network and amplify the overall performance degradation. For instance, in a handover, the Access and Mobility Function (AMF) and the Session Management Function (SMF) coordinate. Then, if the SMF is under a DoS attack, the session modification during handover may be delayed, which may cause a delay in data transmission to the UE.

A. Challenges

Considering these, we focus on three challenges in 5G network security as follows:

- **High vulnerability of the 5G control plane:** The key entry point is the signalling protocols used to control and manage the connectivity. For example, attackers can manipulate messages in the Next Generation Application Protocol (NGAP) to flood resources or alter session states [3]. This can lead to wider disruption to the traditional rule-based mechanisms and prevent meeting the low-latency requirements.
- **Lack of deployment level evaluation:** The attack detection and mitigation mechanisms are primarily evaluated for their accuracy. However, it's necessary to perform a deployment-level testing due to the stringent performance requirements of the 5G/6G networks. Packet captures and their processing at the live network introduce overhead. This is particularly important because these networks will increasingly carry more traffic. On the other hand, testing on the live network can cause disruptions.
- **Limited automation in 5G emulation frameworks:** To analyse and collect data on such threat models, platforms that can replicate complex 5G networks are needed. However, the current emulation frameworks require excessive manual involvement and minimal support for automation. This prevents iterative what-if testing, which in turn hinders the development of accurate and efficient attack detection and mitigation systems.

To address these, Digital Twins (DTs) can be leveraged, which are virtual replicas that are synchronised and replicated environments. These have been utilised to enable monitoring, analysis, prediction, and optimisation [9], [10]. Moreover, they have been explored for anomaly detection, forecasting intrusions, and optimising mitigation strategies [11]. However, most existing studies focus on DTs as a predictive method; however, they are also a key enabler for what-if analysis thanks to the modelling capabilities. This capability has been primarily explored through the use of purely data-driven models, mainly generative AI [12]. However, what-if analysis on secu-

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TABLE I
COMPARISON OF SELECTED RELATED WORKS

Work	RAN	CN	Attack Detection	Automated Orchestration
[4]	✓	×	✓	×
[5]	×	×	×	✓
[6]	✓	×	×	×
[7]	×	✓	×	×
[8]	✓	✓	×	×
Proposed	✓	✓	✓	✓

rity mechanisms also requires explicit modelling for protocol dynamics and attack–response behaviours. For this purpose, simulation approaches have also been used in conjunction; the traffic generation of the scenario is done through data-driven models, and then the simulation is used to perform the what-if analysis [13]. However, the simulation approach can only provide theoretical performance and lacks the runtime accuracy as the software stacks at each node are not considered. For this, an emulation approach has been proposed in [4], where data-driven fuzzing forms attacks on the confidentiality and integrity of the data. However, in this, only the interaction between UE and gNB nodes is considered, but a similar principle can be extended to the control plane.

B. Contributions

Considering these challenges, we propose the Emulation Digital Twin (EDT) Framework to serve as a scalable testing environment, which is shown in Figure 1.

Respectively, we contribute to the literature as follows:

- We present the EDT framework, which consists of three layers. Here, the Emulation Layer utilises Containerlab, UERANSIM, and an extended Open5GS implementation. The Digital Twin (DT) Layer forms a synchronised graph-based representation, and the attack detection and mitigation mechanism is integrated. Moreover, the Orchestration Layer consists of a modular architecture to automate the lifecycle management of scenarios, deployments and experiment generation.
- We extend Open5GS with signalling hooks that capture alarms from processing stages and synchronise them with the DT layer. Then, we integrate attack detection and mitigation modules that operate directly on the live graph-based representation of the network. This enables a closed feedback loop to monitor control-plane behaviour, identify anomalies, execute mitigation strategies, and assess their performance.
- We validate the EDT framework through an NGAP DoS case study and measure the throughput and latency at the data plane. Moreover, we record the lines of code necessary to produce these experiments and measure the resource consumption of the proposed framework compared to traditional approaches. The results show that the proposed framework achieves up to 81% reduction in manual experimentation with only 0.42% computational overhead.

The rest of this paper is organised as follows: Section II discusses the related work and the network emulations in the context of 5G networks. Then Section III presents the EDT framework, including its architecture and what-if analysis methodology. Then the performance evaluations are presented and discussed at Section IV. and the conclusions and future directions at Section V.

II. RELATED WORK

The Denial of Service (DoS) attacks have been highly studied, and numerous approaches have been proposed to detect and mitigate these, from statistical rule-based to machine learning (ML) models. For instance, information entropy has been utilised to filter traffic, followed by the application of a convolutional neural network (CNN) for classification [14]. Similarly, Pearson correlation and Mean Decrease Impurity based feature selection techniques were proposed, and then Random Forest and Long Short-Term Memory (LSTM) are used in [15]. While these approaches are necessary for attack detection at the networking level, the 5G's service-based architecture necessitates a system perspective because the attacks through signalling protocols do not aim to overwhelm the links between nodes but rather aim to deplete resources of the NFs.

In this context, the 5G control plane is vulnerable to attacks on N2 and N4 interfaces through their protocols of NGAP and Packet Forwarding Control Protocol (PFCP). Here, the N2 interface between gNB and AMF is targeted by attacks that focus on User Equipment (UE) registration using NGAP. To address this, in [3], a hybrid ML architecture has been proposed using NGAP telemetry, extracting multi-attribute features and modelling temporal and semantic dependencies. Similarly, the N4 interface between SMF and UPF exposes vulnerabilities in the PFCP protocol by modifying Packet Detection and Forwarding Rules (PDRs/FARs). To address this, a three-based ML model with a game-theoretic algorithm was proposed in [16] to detect session modification attacks through FARs. Similarly, in [17], an LSTM model is used to identify the PFCP signalling floods.

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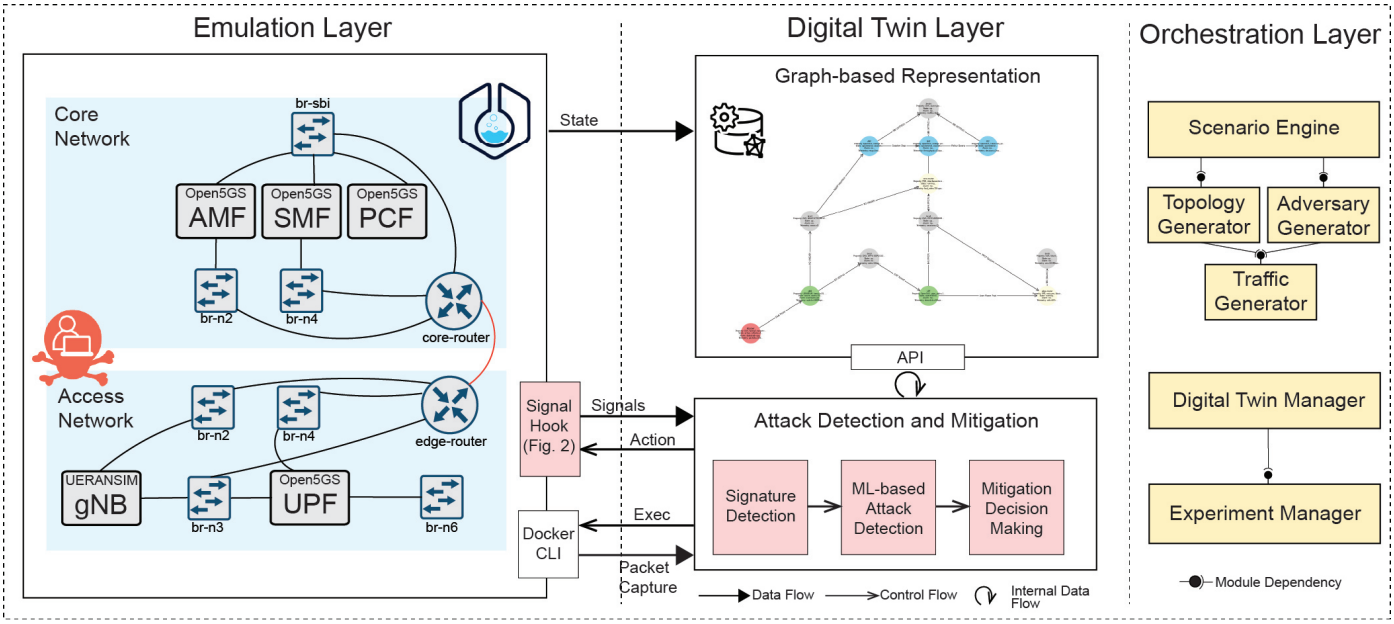


Fig. 1. Emulation Digital Twin (EDT) Framework - Case Study on NGAP Attacks in 5G Control Plane

However, despite the promising results, most of these approaches are evaluated in offline settings with a focus on accuracy and also rely on deep packet inspection, which causes concerns about their detection latency. This limits their applicability in real 5G/6G networks where latency and performance overhead are important for control plane stability and service continuity. Additionally, the 5G security from the perspective of signalling protocols has also been less explored. This limitation can be attributed to the limited availability of data and emulation frameworks [18].

Network emulators such as Mininet, GNS3, and Containerlab enable the execution of networking software with protocol interactions. Here, the Containerlab is part of the recent generation of network emulators, and it is an open-source container-based framework. In this approach, each node is run as a container, forming virtual links, which enables the design of large-scale testing [5]. However, in ContainerLab, the emulation is agnostic to the functionality at each node. Therefore, it can be used to emulate the topology of the 5G/6G network, but the behaviour of each node has to be implemented separately. At this, the OpenAirInterface (OAI) [8], UERANSIM [6], and Open5GS [7] provide a functional implementation of the 3GPP protocol stack. Here, Open5GS is focused on the containerised implementation of the 5G Core, and OAI involves the Radio Access Network (RAN) and core with over-the-air and channel-emulated testing. Moreover, UERANSIM focuses on the protocol-level emulation of the gNBs and UE without including the physical layer. Here, while the OAI provides a full-stack emulation, it introduces a significant computational overhead due to channel emulation [19]. For this purpose, the UERANSIM and Open5GS can be used jointly; however, managing the deployment and various experiments on these emulators is highly complex and resource-intensive. Because each is based on distinct configuration schemas, runtime dependencies, and interfaces [20]. Moreover, for security testing, the experiments have to be modified for topologies which require extensive scripting and manual reconfiguration. As a result, large-scale and iterative security becomes especially challenging for evaluating attack detection and mitigation.

III. Emulation Digital Twin (EDT) Framework

DT capabilities are key for security testing in realistic conditions and can be used to emulate attacks, validating and measuring the performance of detection and mitigation without exposing a live network. Motivated by this, we present the Emulation Digital Twin (EDT) Framework, which is a modular and multi-layered architecture as illustrated in Figure 1.

A. Emulation Layer

This layer forms the network emulation via joint use of ContainerLab, UERANSIM, and our extended implementation of Open5GS. Here, we utilise these for their lightweight capabilities and enable iterative and large-scale security testing. Here, ContainerLab is utilised to manage the deployment of containerised network functions (NFs) in 5G, forming a virtual network. In the scope of this work, we don't focus on the physical layer security and therefore use protocol-level emulation for gNBs and UEs by using UERANSIM. Here, we utilise the Open5GS codebase, which has modular implementations of 3GPP-compliant protocol stacks. However, to enable the DT integration, we extend it with a modular signalling hook shown in Figure 2. Here, these stages are derived from Open5GS's,

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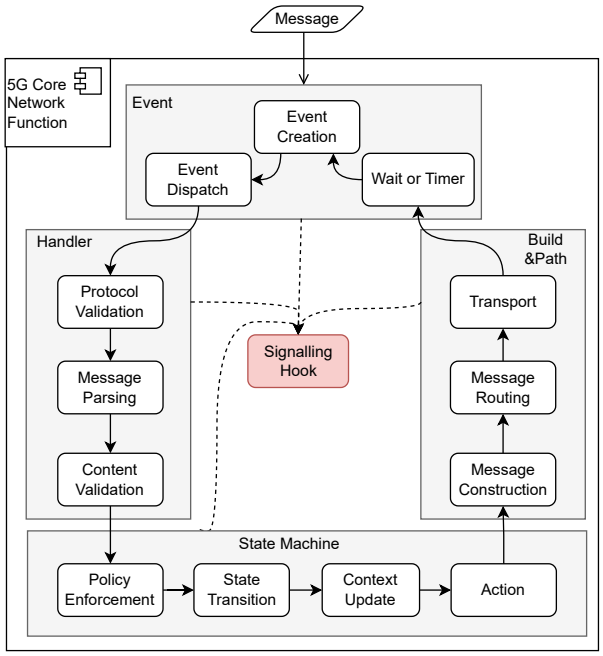


Fig. 2. Integration of the Signalling Hooks at NFs in the context of the Open5GS codebase to Attack Detection and Mitigation

but we note that they are used for the purpose of clarity and to illustrate the modularity exhibited in their codebase; they are not native. Here, at the Message Parsing state, the initial validation of the incoming packets is made. For example, in a DoS attack, the invalid format with random payloads triggers a signal. Moreover, at the State Machine Decision state, the protocol state transitions are monitored to identify anomalies at state transitions, such as authentication bypass attempts. At the Context Update stage, available resources are checked before updating the context. This may correspond to either a legitimate spike due to UE behaviours or to malicious intent. The evaluation of this is performed at the DT Layer.

B. Digital Twin Layer

The DT Layer is built based on the Emulation Layer and includes a descriptive representation, performance monitoring, the attack detection and mitigation modules.

1) Graph-based Representation: The real-time representation of the Emulation Layer is held at the Graph-based Representation Module, and this representation includes the following fields:

- **Property:** This field is used for internal properties and relatively static attributes of each node. For example, these include network slicing configurations, encryption algorithms utilised by the node and the image used for the node.
- **State:** This field is used for the dynamic states of the node, including but not limited to the active load, the container health, and runtime configurations. The updates to this field overwrite the existing value for efficiency in retrieving data from the representation.
- **Alarm:** This field defines the configurations and metadata for the signalling hooks presented at the Emulation Layer.
- **Telemetry:** This field is used to define the metadata for the data streams towards the DT Layer, and it does not store the raw data. This is done for efficiency. Here, these data streams carry high-volume data as packet captures and are sent dynamically based on trigger actions from the IDS. The dynamic triggering of data flows is implemented to prevent continuous monitoring at high data rates.
- **Relationship:** A relationship is an edge formed between two nodes in this graph-based representation. This encodes the contextual information between the nodes and is primarily used for the topology. Moreover, these edges are labelled and do not have attributes for computational efficiency during traversals.

Furthermore, we develop this as a Python module using networkx for the graph and expose a FastAPI interface for integration with other modules.

2) Attack Detection and Mitigation: The attack detection and mitigation are embedded at the DT layer and integrated with the Graph-based Representation module. This working mechanism involves the following steps. (1) Signature Detection monitors the alarms at the graph-based representation. It performs analysis for predefined patterns, including frequency thresholds for floods and sequence matches. When a match is detected, the packet captures for the identified interfaces are initiated. (2) ML-Based Detection uses a classification model to classify the benign and malicious packets. (3) Mitigation Execution retrieves the mitigation strategy defined at the graph-based representation when the attack has been detected and applies it to the Emulation Layer through the signalling hooks.

C. Orchestration Layer

This layer operates at the control plane of the EDT framework, automating the configuration and execution of security tests while managing the Emulation layer with reduced user intervention. This allows the experimentation to be automated, enabling the application of large-scale and iterative security tests.

- **Scenario Engine:** This can be considered as the central or root point of this layer. The scenario engine is used to define the experimentation parameters, such as the topology, NF placements, attack configurations, benign traffic patterns, and the integration of the attack detection and mitigation applications. For instance, given the configuration file, the scenario engine performs a multi-phase test for baseline performance, attack injection, mitigation activation, and post-analysis. Additionally, the schema that

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will be used by the graph-based representation and the models for the attack detection and mitigation are defined by the scenario engine.

- **Twin Manager:** The configuration and management of the DT layer is done through this model. This integrates with the Experiment Manager to configure the signalling hooks at the Emulation layer. This provisions the synchronisation between the DT layer and the Emulation layer.
- **Topology Generator:** This module is managed by the Scenario Engine and performs the deployment of the topology at the Containerlab. For this purpose, it generates YAML files and container configurations that include container images, virtual links, and link impairments affecting latency, jitter, and packet loss.
- **Traffic Generator:** This module is also managed by the Scenario Engine and used to simulate the benign traffic. Here, it receives the configurations for the traffic to be applied and scripts benign flows for different kinds of application traffic. This is automated through the templates implemented, which can be configured for parameters such as packet size and burst behaviour.
- **Attack Generator:** This module operates in coordination with the Scenario Engine and Traffic Generator. For example, data-plane overload in GTP tunnel flooding, and configuration tampering to alter policies at 5GC are generated through the Attack Generator.
- **Experiment Manager:** This module is responsible for the lifecycle of the experiments and the topologies deployed using container lab. Here, the QoS of the UEs, the key performance indicators of the NFs, experiment status, deployment health, and the experiment resource consumption are configured to be recorded and sent to DT.

IV. Case Study and Results

In this article, we consider the DoS attack over NGAP. This attack targets the UE registration procedure at the AMF by sending high-volume packets with corrupted UE-NGAP-ID. This leads to an error condition at the Handler, or if it passes, the resources are allocated at the State Machine stage, causing resource exhaustion in the AMF. Therefore, we configure the signalling hook to record these. Then, the ML-based detection is done through the proposed method in the [3], which is based on a hybrid neural network where a Temporal Convolution Network (TCN) is used for the flow sequence and an MLP is used to model the semantic features at the Information Element (IE) in the message and these are connected with dense layers for the attack classification.

To evaluate the performance, we measure the performance under four strategies for attack detection and mitigation. This includes, without the DT layer active under benign traffic only, both benign traffic and attacks, for both with and without the DT layer active. During these measures, the throughput and latency of the data plane of the 5G network are considered.

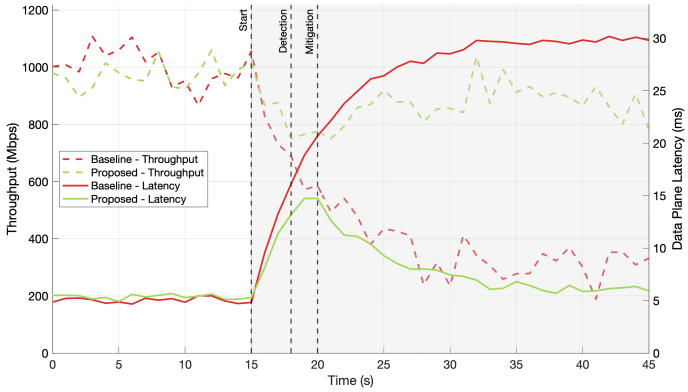


Fig. 3. Comparison of Throughput and Data Plane Latency during DoS Attack

Moreover, we measure the resource utilisation of the deployed machine in terms of CPU and RAM. Here, the multi-stage testing and measurement process is automated as described in Section III-C.

In the Figure 3, we show the data plane latency and throughput over the experimentation timeline. Here, the baseline consists of no attack detection mechanism being employed. Here, the attack is initiated at 15s, resulting in a decrease in the benign users' throughput and an increase in their latency. The start of the attack triggers the signalling hooks to be triggered as explained in Section III-A. With this, the DT layer identifies the attack and retrieves a small-time window packet capture to validate this signature. Then, according to the decision, the mitigation strategy is initiated to blacklist the attacker node from the network. With this, the latency and throughput return to normal. However, we also observe that the QoS does not recover to the previous case. This is because, while the attacker is blacklisted, they continue to generate traffic and still consume a small portion of resources before it's dropped.

Then, we measure the CPU consumption and total lines of code for these experiments as shown in the Figure 4. Here, the average CPU usage measurement has been averaged for the number of cores in the deployed machine, which has an AMD Ryzen 9 7900X with 12 cores. Here, the comparison is made between pure Open5GS, Open5GS + ContainerLab, and Open5GS + ContainerLab + EDT. Respectively, we observe that the CPU usage remains relatively consistent. This is because a significant part of the computing load comes from the emulated nodes. Therefore, in a pure Open5GS case, the scripting required for CLI and configuration results in the highest number of lines of code. On the other hand, ContainerLab simplifies deployment by defining it through YAML and then handling the deployment; however, for each experiment, the traffic configurations have to be made manually. Here, the proposed framework enables most of these processes to be automated, resulting in an overall reduction of 81%. This shows the proposed framework's performance on enabling large-scale security testing.

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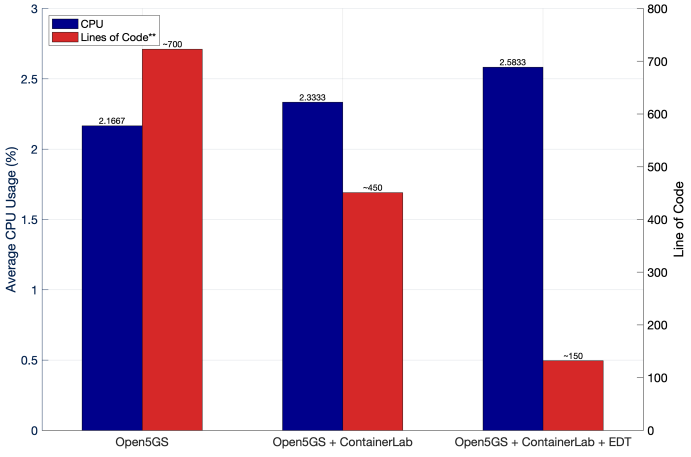


Fig. 4. Comparison of CPU consumption and total Line of Code for this paper's experiments

V. Conclusion

In this work, we address the 5G network security challenges in vulnerability of the control plane, the lack of deployment-level evaluation, and the limited automation of 5G emulation frameworks. For this, we present a modular and layered framework named the Emulation Digital Twin (EDT) Framework. In this, at the Emulation Layer, we utilise Containerlab, UERANSIM, and Open5GS. Here, we extend the Open5GS with the developed signalling hook to retrieve alarms to the Digital Twin (DT) layer. Then, at the DT Layer, we maintain a synchronised graph-based representation and integrate the attack detection and mitigation modules. Meanwhile, the Orchestration Layer automates the configuration, deployment, and lifecycle of the scenarios and experiments. This enables iterative and large-scale what-if analyses with minimal manual intervention. Furthermore, we evaluate the performance under a Next Generation Application Protocol (NGAP) Denial-of-Service (DoS) case study. The results show that the EDT framework enables automated configuration, which enables up to 81% line of code reduction in experimentation setups. As future work, we aim to extend EDT to support distributed emulation and AI-driven scenario generation, thereby enabling a fully autonomous emulation framework.

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